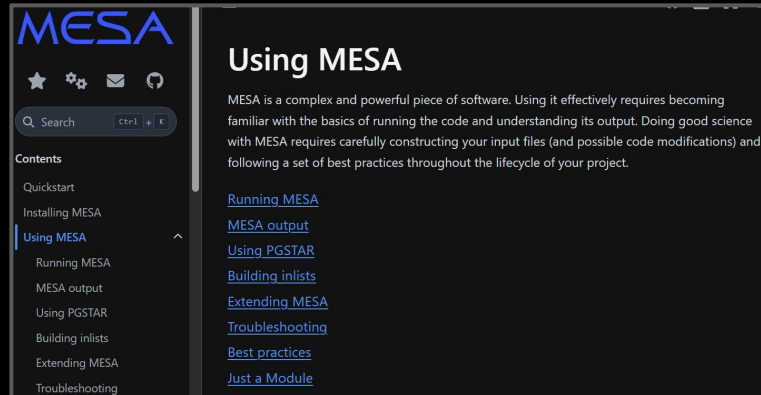


Using MESA

By: Sherman Losey

MESA: Resources

- [Documentation](#)
- [How to install?](#)
- [MESA Summer School 2026 Labs](#)
 - [My summary document](#)
 - [My drive with slides and lab resources](#)



MESA: QOL Setup

- vim ~/.bashrc

```
function mesa-26.04.1 {  
  export MESA_DIR=/home/sherman_losey/mesa-26.04.1  
  export MESASDK_ROOT=/home/sherman_losey/mesasdk  
  source $MESASDK_ROOT/bin/mesasdk_init.sh  
  export OMP_NUM_THREADS=8  
}
```

- mesa-26.04.1 (must use function every new terminal)

```
(base) sherman_losey@Shermans-Laptop:~$ mesa-26.04.1  
Version mesa-26.04.1  
OMP_NUM_THREADS set to 8
```

```
(base) sherman_losey@Shermans-Laptop:~$ echo $OMP_NUM_THREADS  
8
```

MESA: Templates

- Lonely Star:
 - `cp -r ~/mesa-26.04.1/star/work/ .`
- Binary System:
 - `cp -r ~/mesa-26.04.1/binary/work/ .`
- Test Suites
 - Starter kits but not recommended for real science.
 - `cd ~/mesa-26.04.1/star/test_suite`

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/star/test_suite$ ls
1.3M_ms_high_Z      check_pulse_atm      gyre_in_mesa_wd      make_zams_ultra_high_mass  split_burn_big_net
1.4M_ms_op_mono      check_redo            hb_2M                 ns_c                      starspots
1.5M_with_diffusion  conductive_flame      high_mass             ns_h                     test_case_template
12M_pre_ms_to_core_collapse  conserve_angular_momentum  high_rot_darkening  ns_he                   test_memory
15M_dynamo           conv_core_cpm        high_z               other_physics_hooks     test_suite_helpers
16M_conv_premix      count_tests          hse_riemann         pisp                     timing
16M_predictive_mix   custom_colors        irradiated_planet    ppisp                   twin_studies
1M_pre_ms_to_wd      custom_rates         list_tests           radiative_levitation    tzo
1M_thermohaline      debugging_stuff_for_inlists  low_z               relax_composition_j_entropy  update_docs_figures
20M_pre_ms_to_core_collapse  diffusion_smoothness  magnetic_braking     report                  wd_acc_small_dm
20M_z2m2_high_rotation  dol_rsp_test_source   make_brown_dwarf     rsp_BEP                 wd_aic
5M_cepheid_blue_loop   dol_test_source       make_co_wd           rsp_B LAP               wd_c_core_ignition
7M_premsto_AGB         each_test_clean       make_env             rsp_Cepheid             wd_cool_0.6M
R_CrB_star            each_test_compile     make_he_wd           rsp_Cepheid_6M          wd_diffusion
T_tau_gradr           each_test_do          make_metals          rsp_Delta_Scuti        wd_he_shell_ignition
accreted_material_j   each_test_run         make_o_ne_wd         rsp_RR_Lyrae            wd_nova_burst
adjust_net            extended_convective_penetration  make_planets        rsp_Type_II_Cepheid    wd_stable_h_burn
build_and_run         gyre_in_mesa_bcep     make_pre_ccsn_13bvn  rsp_check_2nd_crossing  zams_to_cc_80
c13_pocket            gyre_in_mesa_envelope  make_sdb             rsp_gyre                simplex_solar_calibration
carbon_kh             gyre_in_mesa_ms       make_zams            rsp_save_and_load_file
cburn_inward          gyre_in_mesa_rsg      make_zams_low_mass   semiconvection
ccsn_IIP              gyre_in_mesa_spb      make_zams_low_mass   simplex_solar_calibration
```

MESA: Speedrun

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls
LOGS  README.rst  clean  inlist  inlist_pgstar  inlist_project  make  mk  photos  plot.py  re  rn  src  star
```

- **Inlists:**
 - inlist: Set what inlists we call.
 - inlist_project: Set starting/ending properties of the star like mass, temp, metallicity, etc.
 - inlist_pgstar: Set up plotting program.
 - etc.
- **LOGS**
 - history.data: Experiment outputs like model #, mass, Teff, Luminosity, etc.
- **src/run_star_ectras.f90: Code the stellar processes/interactions**
 - Accretion, convection, magnetic braking, energy/mass loss/gain, etc.
- **Misc**
 - png (or whatever you name it): pgstar plot outputs
 - photos: binary files to continue a run

MESA: Defaults

`&star_job`

`! see star/defaults/star_job.defaults`

`&controls`

`! see star/defaults/controls.defaults`

`&pgstar`

`! see star/defaults/pgstar.defaults`

- `ls ~/mesa-26.04.1/star/defaults`

```
(base) sherman_losey@Shermans-Laptop:~$ ls ~/mesa-26.04.1/star/defaults
FORMAT                controls_dev.defaults  pgstar.README        profile_columns.list  star_job_dev.defaults
controls.defaults     history_columns.list   pgstar.defaults      star_job.defaults
(base) sherman_losey@Shermans-Laptop:~$ |
```

A Simple MESA Model

- Straight from the work directory
 - `cp -r ~/mesa-26.04.1/star/work/ .`
- Properties
 - Starting Mass = $15M_{\odot}$
 - $z = 0.02$
- Evolution
 - Protostar $\rightarrow$ ZAMS
 - ZAMS $\rightarrow$ RGB

Simple MESA Run: inlist

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls  
LOGS  README.rst  clean  inlist  inlist_pgstar  inlist_project  make  mk  photos  plot.py  re  rn  src  star
```

```
&star_job  
  
    read_extra_star_job_inlist(1) = .true.  
    extra_star_job_inlist_name(1) = 'inlist_project'  
  
/ ! end of star_job namelist  
  
&eos  
  
    read_extra_eos_inlist(1) = .true.  
    extra_eos_inlist_name(1) = 'inlist_project'  
  
/ ! end of eos namelist  
  
&kap  
  
    read_extra_kap_inlist(1) = .true.  
    extra_kap_inlist_name(1) = 'inlist_project'  
  
/ ! end of kap namelist  
  
&controls  
  
    read_extra_controls_inlist(1) = .true.  
    extra_controls_inlist_name(1) = 'inlist_project'  
  
/ ! end of controls namelist
```

&pgstar

```
    read_extra_pgstar_inlist(1) = .true.  
    extra_pgstar_inlist_name(1) = 'inlist_pgstar'
```

Simple MESA Run: inlist_project

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls
LOGS  README.rst  clean  inlist  inlist_pgstar  inlist_project  make  mk  photos  plot.py  re  rn  src  star
```

&star_job

! see star/defaults/star_job.defaults

! begin with a pre-main sequence model
create_pre_main_sequence_model = .true.

! save a model at the end of the run
save_model_when_terminate = .true.
save_model_filename = '15M_at_TAMS.mod'

! display on-screen plots
pgstar_flag = .true.

/ ! end of star_job namelist

&controls

! see star/defaults/controls.defaults

! starting specifications

initial_mass = 15 ! in Msun units
initial_z = 0.02

! when to stop

! stop when the star nears ZAMS ($L_{\text{nuc}}/L > 0.99$)
Lnuc_div_L_zams_limit = 0.99d0
stop_near_zams = .true.

! stop when the center mass fraction of h1 drops below this limit
xa_central_lower_limit_species(1) = 'h1'
xa_central_lower_limit(1) = 1d-3

! wind

! atmosphere

! rotation

! element diffusion

Simple MESA Run: inlist_pgstar

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls  
LOGS README.rst clean inlist inlist_pgstar inlist_project make mk photos plot.py re rn src star
```

```
!pgstar
```

```
! see star/defaults/pgstar.defaults  
  
! MESA uses PGPLOT for live plotting and gives the user a tremendous  
! amount of control of the presentation of the information.
```

```
! show HR diagram  
! this plots the history of L, Teff over many timesteps  
HR_win_flag = .true.
```

```
! set static plot bounds  
HR_logT_min = 3.5  
HR_logT_max = 4.6  
HR_logL_min = 2.0  
HR_logL_max = 6.0
```

```
! set window size (aspect_ratio = height/width)  
HR_win_width = 6  
HR_win_aspect_ratio = 1.0
```

```
! save HR plot  
HR_file_flag = .true.  
HR_file_dir = 'png'  
HR_file_prefix = 'hr_'  
HR_file_interval = 10
```

```
! show temperature/density profile  
! this plots the internal structure at single timestep  
TRho_Profile_win_flag = .true.
```

```
! add legend explaining colors  
show_TRho_Profile_legend = .true.
```

```
! display numerical info about the star  
show_TRho_Profile_text_info = .true.
```

```
! set window size (aspect_ratio = height/width)  
TRho_Profile_win_width = 8  
TRho_Profile_win_aspect_ratio = 0.75
```

```
! save TRho_Profile  
TRho_Profile_file_flag = .true.  
TRho_Profile_file_dir = 'png'  
TRho_Profile_file_prefix = 'trho_'  
TRho_Profile_file_interval = 10
```

```
end of pgstar namelist
```

Simple MESA Run: run_star_extras.f90

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls
LOGS  README.rst  clean  inlist  inlist_pgstar  inlist_project  make  mk  photos  plot.py  re  rn  src  star
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/work$ ls src/
run.f90  run_star_extras.f90
```

```
module run_star_extras

    use star_lib
    use star_def
    use const_def
    use math_lib

    implicit none

    ! these routines are called by the standard run_star check_model
    contains

    include 'standard_run_star_extras.inc'

end module run_star_extras
```

Simple MESA Run: Starting

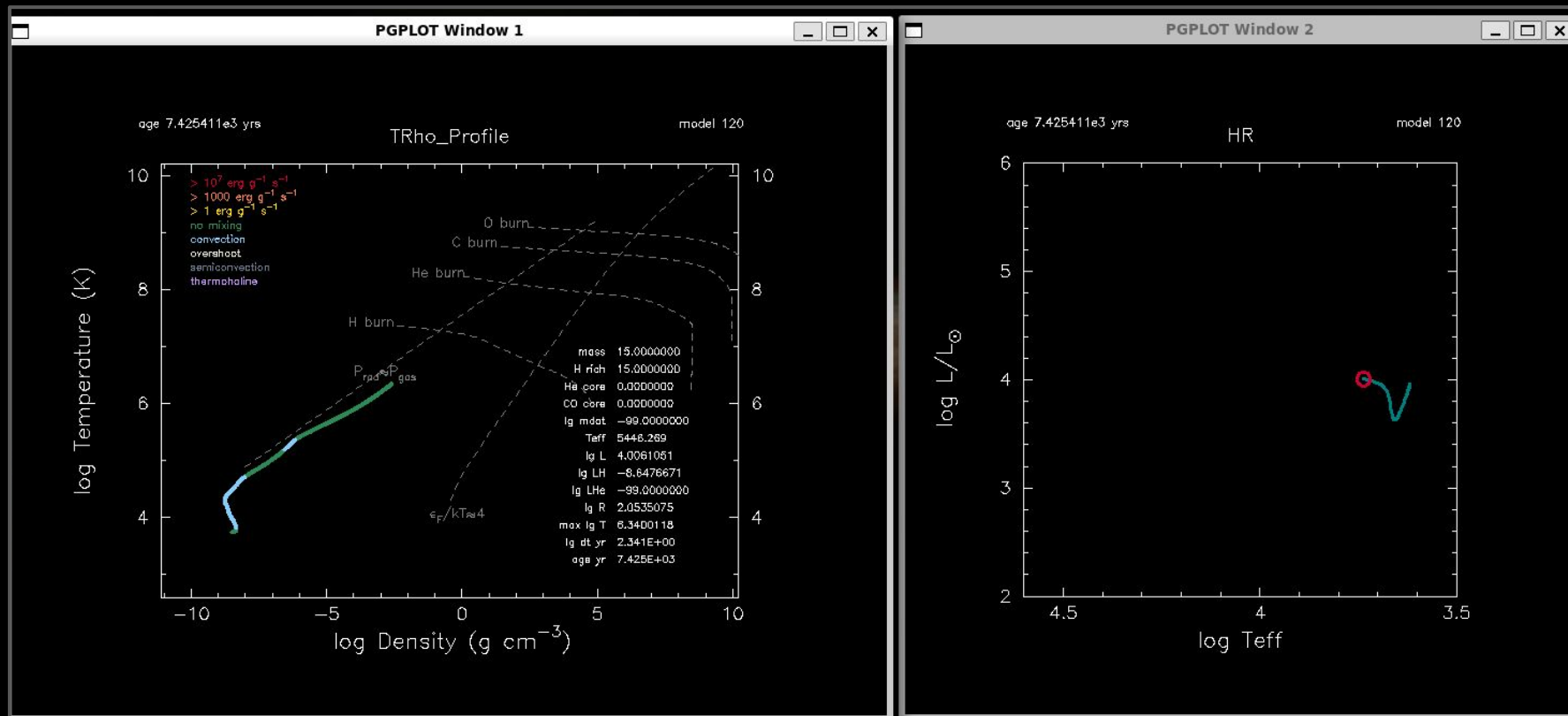
Always `./mk` every time you edit `run_star_extras.f90`

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/Example$ ./clean
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/Example$ ./mk
gfortran -Wno-uninitialized -fno-range-check -fmax-errors=7 -fdiagnostics-color -fprotect-parens -fno-sign-zero -fbacktrace -ggdb -finit-real=snan -fopenmp -fbounds-check -Wuninitialized -Warray-bounds -ggdb -ffree-form -ffree-line-length-none -x f95-cpp-input -std=f2008 -Wno-error-tabs -I/home/sherman_losey/mesa-26.04.1/include -I../src -c ../src/run_star_extras.f90
gfortran -Wno-uninitialized -fno-range-check -fmax-errors=7 -fdiagnostics-color -fprotect-parens -fno-sign-zero -fbacktrace -ggdb -finit-real=snan -fopenmp -fbounds-check -Wuninitialized -Warray-bounds -ggdb -ffree-form -ffree-line-length-none -x f95-cpp-input -std=f2008 -Wno-error-tabs -I/home/sherman_losey/mesa-26.04.1/include -I../src -c /home/sherman_losey/mesa-26.04.1/star/job/run_star.f90
gfortran -Wno-uninitialized -fno-range-check -fmax-errors=7 -fdiagnostics-color -fprotect-parens -fno-sign-zero -fbacktrace -ggdb -finit-real=snan -fopenmp -fbounds-check -Wuninitialized -Warray-bounds -ggdb -ffree-form -ffree-line-length-none -x f95-cpp-input -std=f2008 -Wno-error-tabs -I/home/sherman_losey/mesa-26.04.1/include -I../src -c ../src/run.f90
gfortran -fopenmp -o ../star run_star_extras.o run_star.o run.o -L/home/sherman_losey/mesa-26.04.1/lib -lstar -lstar_data -lgyre_mesa -lgyre -lionization -latm -lcolors -lturb -lnet -leos -lkap -lrates -lneu -lchem -linterp_2d -linterp_1d -lnum -lauto_diff -lforum -lmtx -lconstr -lmath -lutils 'mesasdk_crmath_link' 'mesasdk_lapack95_link' 'mesasdk_lapack_link' 'mesasdk_blas_link' 'mesasdk_hdf5_link' 'mesasdk_pgplot_link' -lz
/home/sherman_losey/mesasdk/bin/./lib/gcc/x86_64-pc-linux-gnu/13.3.0/../../../../x86_64-pc-linux-gnu/bin/ld: warning: atm_support.o: requires executable stack (because the .note.GNU-stack section is executable)
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/Example$ ./rn
DATE: 2026-06-24
TIME: 10:28:58
reading user weak rate file /home/sherman_losey/mesa-26.04.1/data/rates_data/rate_tables/S13_r_be7_wk_li7.h5
version_number r26.4.1
WARNING: Failed to find colors namelist in file: inlist
read inlist_project
```

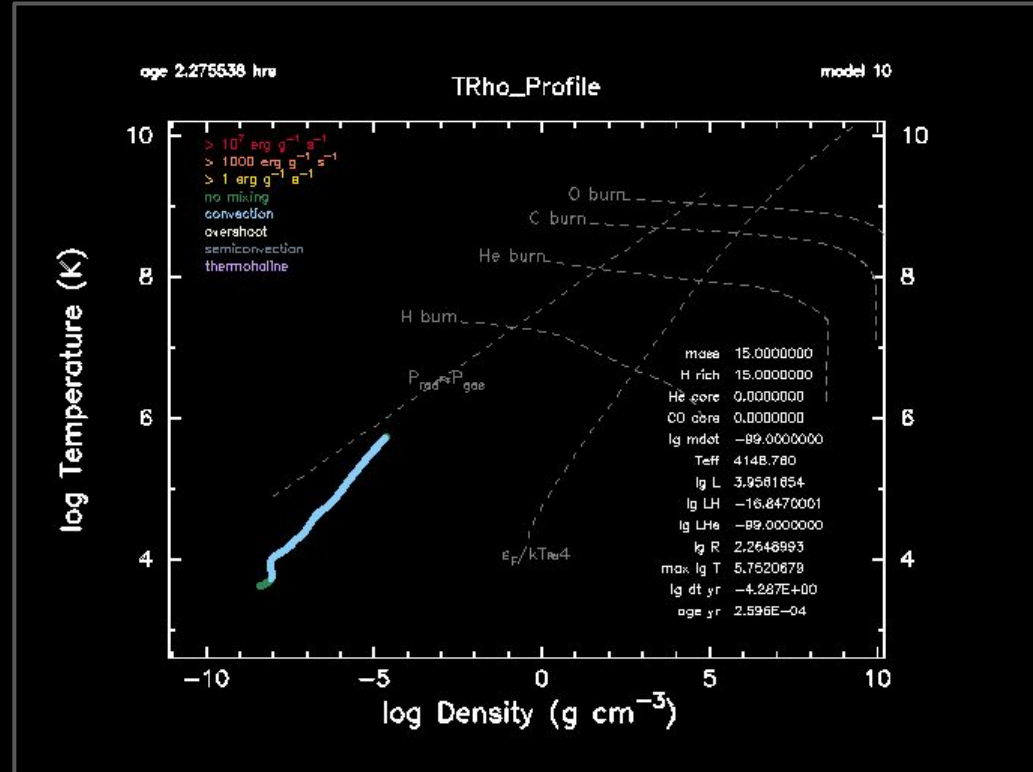
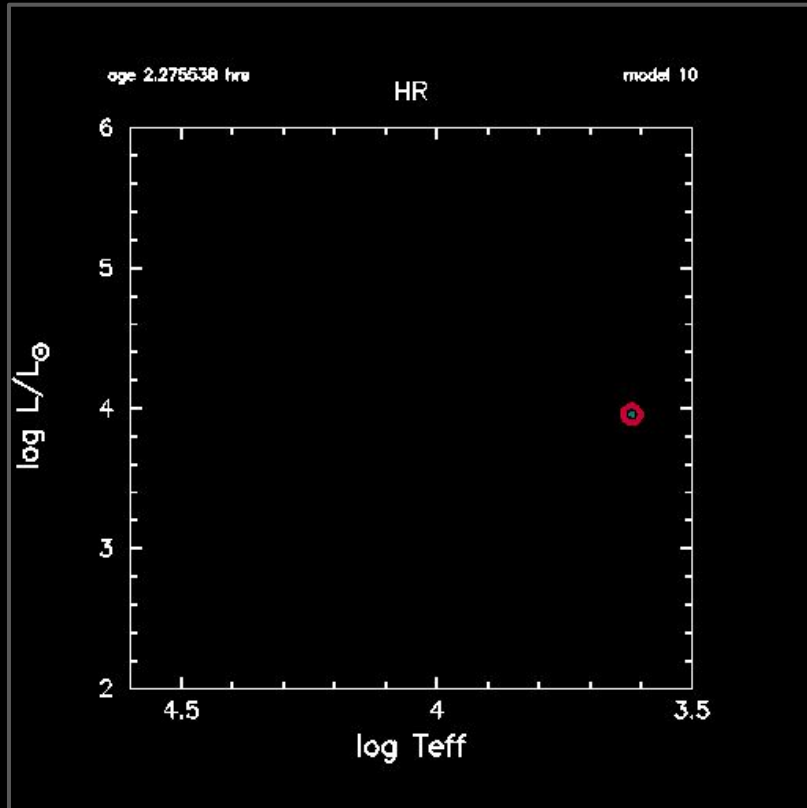
The terminal output contains the following information

```
'step' is the number of steps since the start of the run,
'lg_dt' is log10 timestep in years,
'age_yr' is the simulated years since the start run,
'lg_Tcntr' is log10 center temperature (K),
'lg_Dcntr' is log10 center density (g/cm^3),
'lg_Tmax' is log10 max temperature (K),
'Teff' is the surface temperature (K),
'lg_R' is log10 surface radius (Rsun),
'lg_L' is log10 surface luminosity (Lsun),
```

Simple MESA Run: Plot



Simple MESA Run: GIF of Run



Simple MESA Run: Stopping Condition

- Comment out stopping conditions with a !

```
&controls

! see star/defaults/controls.defaults

! starting specifications
initial_mass = 15 ! in Msun units
initial_z = 0.02

! when to stop

! stop when the star nears ZAMS ( $L_{\text{nuc}}/L > 0.99$ )
!Lnuc_div_L_zams_limit = 0.99d0
!stop_near_zams = .true.

! stop when the center mass fraction of h1 drops below this limit
!xa_central_lower_limit_species(1) = 'h1'
!xa_central_lower_limit(1) = 1d-3
```

Simple MESA Run: Resuming

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/Example$ ./re
```

```
restart from x207
```

```
DATE: 2026-06-24
```

```
TIME: 11:02:19
```

```
reading user weak rate file /home/sherman_losey/mesa-26.04.1/data/rates_data/rate_tables/S13_r_be7_wk_li7.h5
```

```
version_number r26.4.1
```

```
WARNING: Failed to find colors namelist in file: inlist
```

```
read inlist_project
```

```
read restart_photo
```

```
net name basic.net
```

```
kap_option gs98
```

```
kap_CO_option gs98_co
```

```
kap_lowT_option lowT_fa05_gs98
```

```
OMP_NUM_THREADS
```

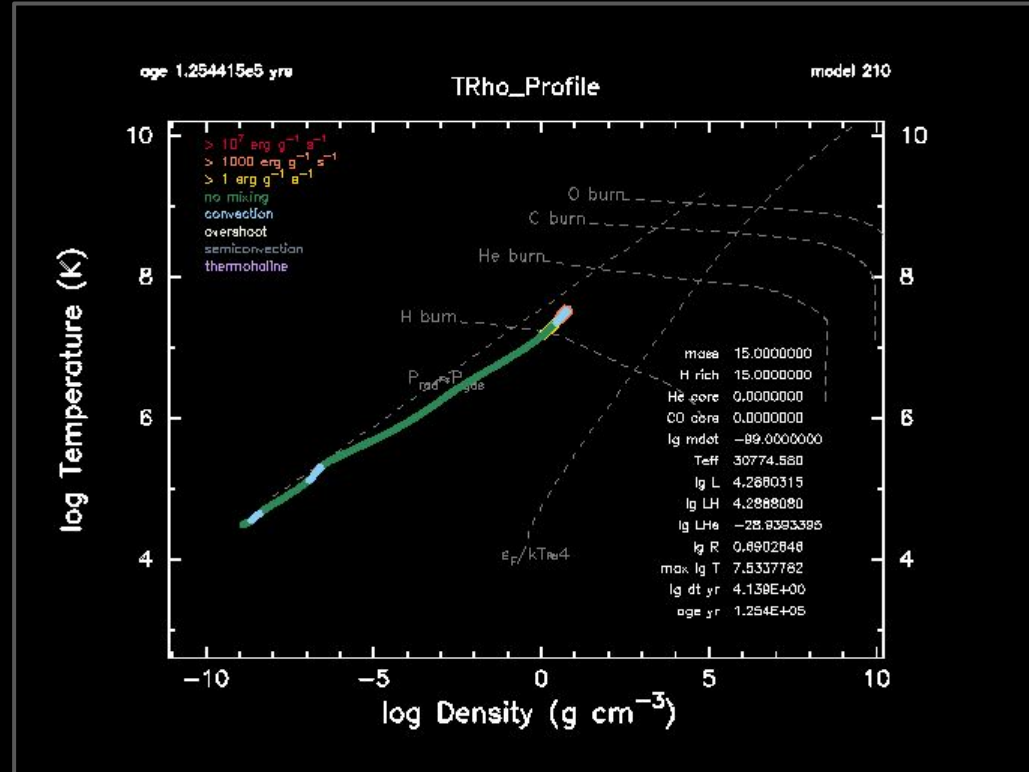
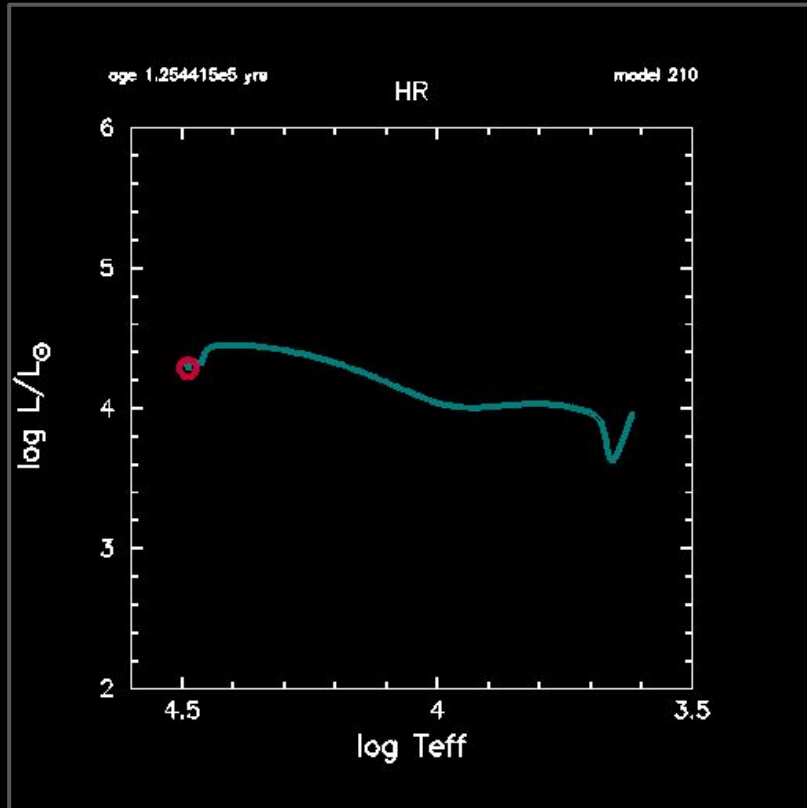
```
8
```

step	lg_Tmax	Teff	lg_LH	lg_Lnuc_tot	Mass	H_rich	H_cntr	N_cntr	Y_surf	eta_cntr	zones	retry
lg_dt_yrs	lg_Tcntr	lg_R	lg_L3a	lg_Lneu	lg_Mdot	He_core	He_cntr	O_cntr	Z_surf	gam_cntr	iters	
age_yrs	lg_Dcntr	lg_L	lg_LZ	lg_Lphoto	lg_Dsurf	CO_core	C_cntr	Ne_cntr	Z_cntr	v_div_cs	dt_limit	

208	7.533494	3.077E+04	4.287582	4.287582	15.000000	15.000000	0.698179	0.005140	0.280000	-5.738059	914	3
-----	----------	-----------	----------	----------	-----------	-----------	----------	----------	----------	-----------	-----	---

Runs the model from last
binary photo

Simple MESA Run: GIF of Continuation



Pulsators in MESA

By: Sherman Losey

GYRE

- GYRE is a tool for calculating eigenvalues for a given models oscillatory modes (p (pressure) and g (gravity) modes).
- It essentially finds what modes could be supported based on the models properties.
- Follow the instructions [here](#) to install GYRE

```
function mesa-26.04.1 {  
export MESA_DIR=/home/sherman_losey/mesa-26.04.1  
export MESASDK_ROOT=/home/sherman_losey/mesasdk  
source $MESASDK_ROOT/bin/mesasdk_init.sh  
export GYRE_DIR=$MESA_DIR/gyre/gyre  
export OMP_NUM_THREADS=8  
echo "Version mesa-26.04.1"  
echo "OMP_NUM_THREADS set to 8"  
echo "GYRE Version 8.1"  
}
```

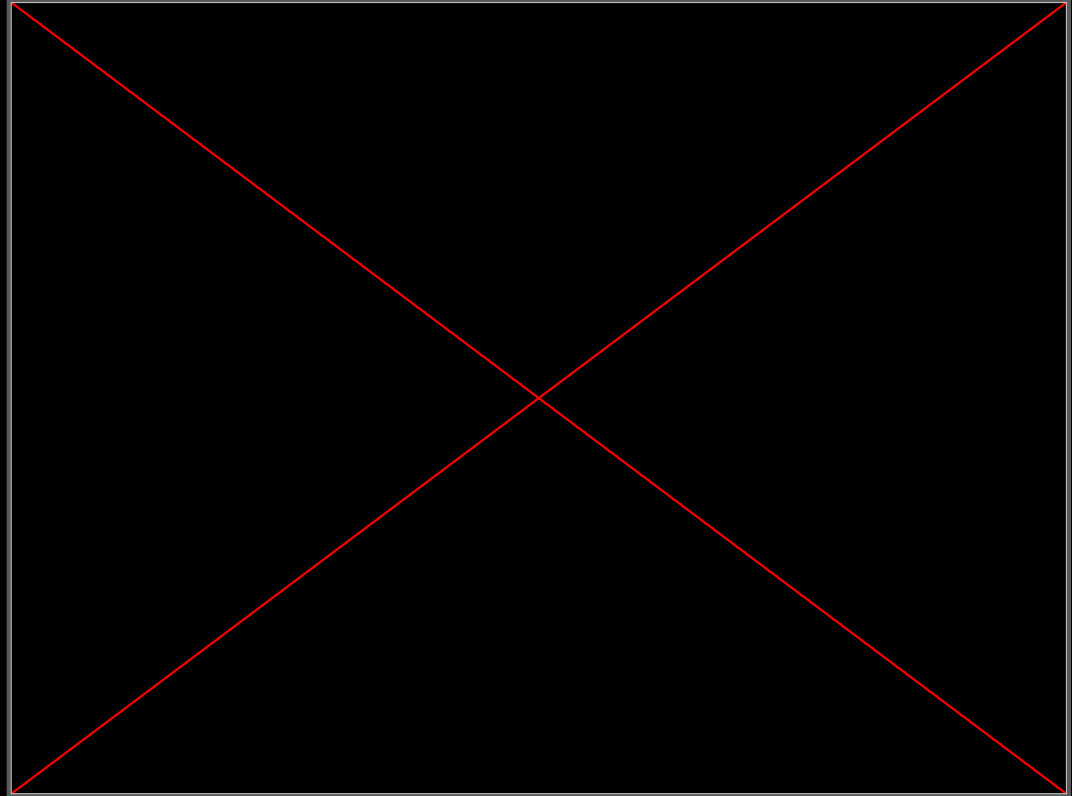
Pulsators: rsp test_suites

- [rsp_Cephied](#)



- [rsp_Delta_Scuti](#)

- [rsp_RR_Lyrae](#)

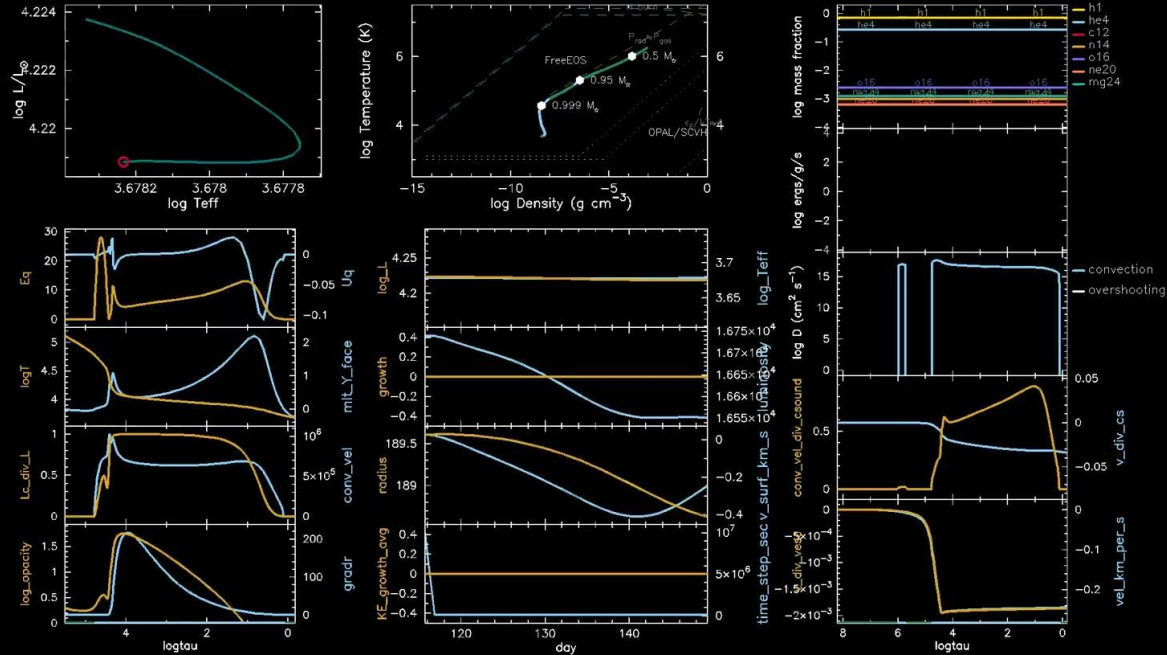


Pulsators: MESA School 2026 Labs

- Tuesday
 - Using GRYRE to model stellar pulsation and then seeing how a chemical composition gradient and/or convective overshoot would effect the model.
- Friday
 - Evolving a Cepheid, giving it nonlinear modes, and identifying bumps on its waveform.

Pulsators: Friday Lab 3

model_number	30	log_dt	-2.4991040	Mass	8.5287664	H_cnr	0.7209867	H_rich	8.5287664
non_fe_core_infall	0	star_age	0.4087763	lg_Mdot	-99.0000000	He_cnr	0.2730419	He_core	0
KE_growth_avg	0	growth	0	eta_cnr	-10.1593291	C_cnr	6.437E-04	CO_core	0
period	0	log_LH	-99.0000000	lg_Lnuc_tot	-99.0000000	O_cnr	0.0025163	radius	188.6310709
num_periods	0	log_LHe	-99.0000000	lg_Lneu	-99.0000000	Ne_cnr	6.194E-04	zones	180
log_cnr_Rho	-3.0414208	log_LZ	-99.0000000	lg_Lphoto	-99.0000000	retries	0	log_cnr_T	6.2654212



Binaries in MESA

By: Sherman Losey

Binaries: Process

- MESA simulates two stars through their lives simultaneously
 - Star 1 → Hydrostatic Evolution (unless you add anything)
 - Star 2 → Hydrostatic Evolution (unless you add anything)
- Each timestep MESA checks the interactions and changes the following accordingly
 - Angular Momentum loss/gain ($\dot{J}$)
 - Eccentricity
 - Mass transfer/loss ($\dot{M}$)
 - Tides
- The stars can have different
 - Masses
 - Evolution Stages
 - Rotation Periods
 - etc.

Binaries: Binary test_suites

~/binary/test_suite Directory

```
(base) sherman_losey@Shermans-Laptop:~$ ls mesa-26.04.1/binary/test_suite/
count_tests      each_test_clean  evolve_both_stars  jdot_ml_check    star_plus_point_mass
dol_test_source  each_test_compile  jdot_gr_check      list_tests        star_plus_point_mass_explicit_mdots
double_bh        each_test_run     jdot_ls_check      report            wind_fed_bhhmxb
```

Directory before the run

```
(base) sherman_losey@Shermans-Laptop:~$ ls mesa-26.04.1/binary/test_suite/evolve_both_stars/
README.rst  clean  history_columns.list  inlist1  inlist_pgbinary  inlist_project  mk  re  rn1
ck          docs  inlist               inlist2  inlist_pgstar    make            profile_columns.list  rn  src
(base) sherman_losey@Shermans-Laptop:~$ |
```

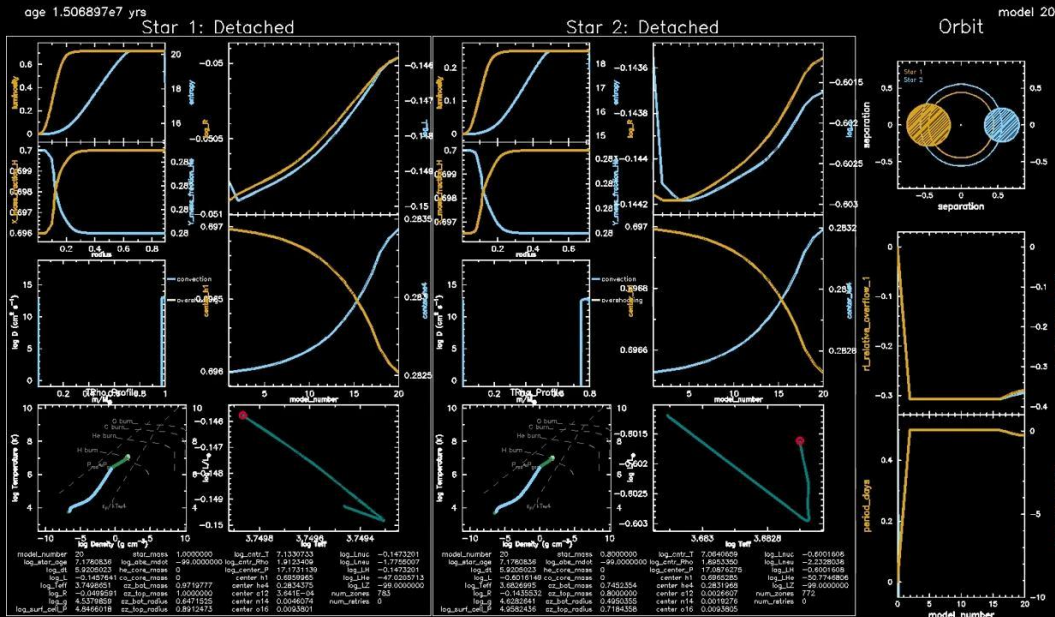
Directory after the run

```
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/show_mesa/test_evolve_both_stars$ ls
LOGS1  binary  clean  inlist  inlist_pgbinary  log1  mk  profile_columns.list  rn1
LOGS2  binary_history.data  docs  inlist1  inlist_pgstar  log2  photos  re  src
README.rst  ck  history_columns.list  inlist2  inlist_project  make  png  rn  testhub.yml
(base) sherman_losey@Shermans-Laptop:~/mesa-26.04.1/projects/show_mesa/test_evolve_both_stars$ |
```

Binaries: Binary test_suites

```
(base) sherman_losey@Shermans-Laptop:~$ ls mesa-26.04.1/binary/test_suite/
```

count_tests	each_test_clean	evolve_both_stars	jdot_ml_check	star_plus_point_mass
dol_test_source	each_test_compile	jdot_gr_check	list_tests	star_plus_point_mass_explicit_mdots
double_bh	each_test_run	jdot_ls_check	report	wind_fed_bhhmxb



Binaries: MESA School 2026 Labs

- Thursday
 - Lab 1: Mass transfer
 - Lab 2: Rapid rotation
 - Lab 3: Evolving into a double BH binary

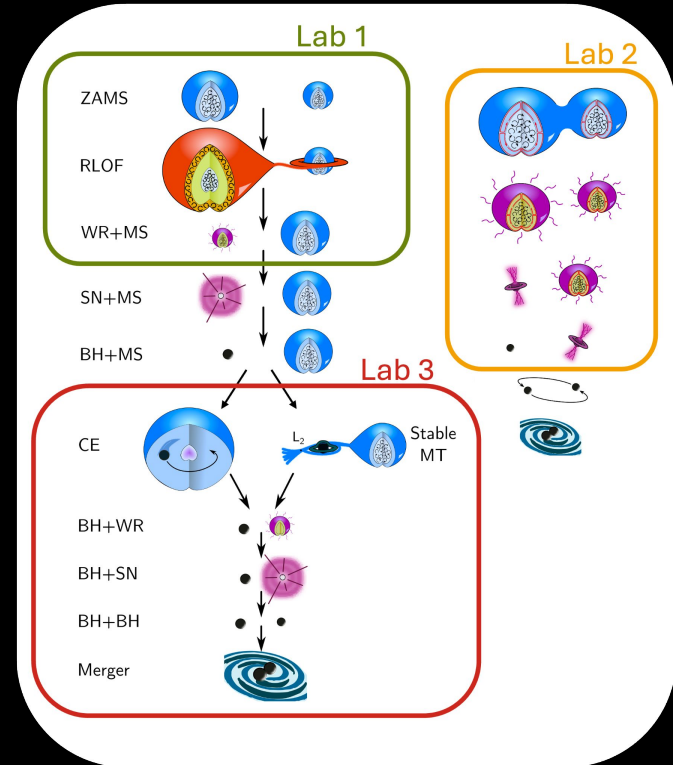
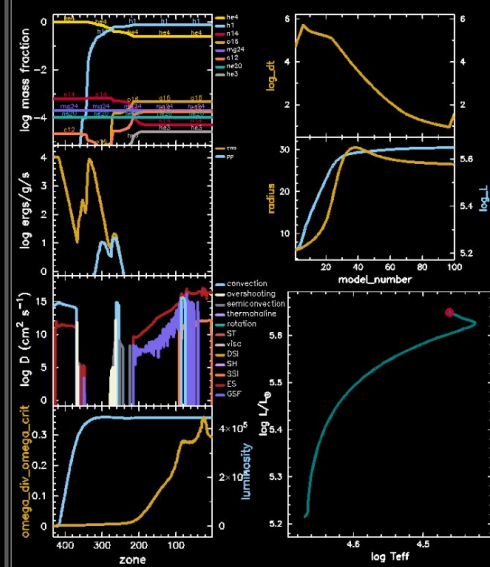


Image Source: [Summer 2026 MESA Lab](#)

Binaries: Thursday Lab 1

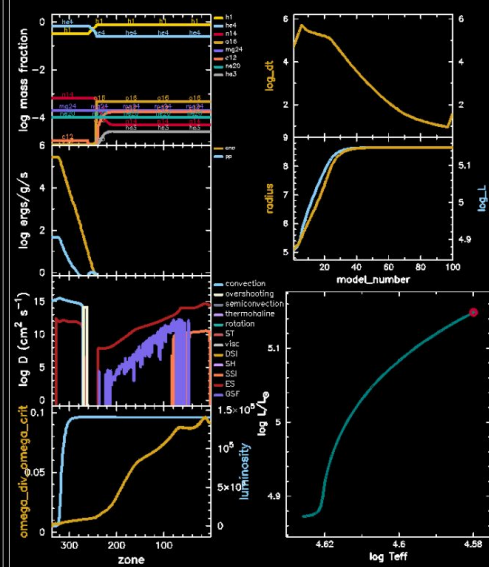
age 5.751259e6 yrs

Star 1: Detached



model_number	100	log_obs_mdott	-6.6551935	star_mass	34.3111161
log_dt	1.5740304	he_core_mass	16.6562063	log_Lnuc	4.1616956
log_L	5.6492894	co_core_mass	0	log_Lneu	2.9966215
log_Teff	4.4622290	center_h1	3.491E-07	log_LH	4.1616956
log_R	1.4226398	center_he4	0.9990327	log_LHe	-5.8599521
log_g	3.1282228	kh_timescale	2.349E+03	num_zones	432
surface_h1	0.7511996	mdot_timescale	1.551E+08	num_retries	0
surface_he4	0.2477739	surf_avg_omega_div_omega_crit	0.2882193		

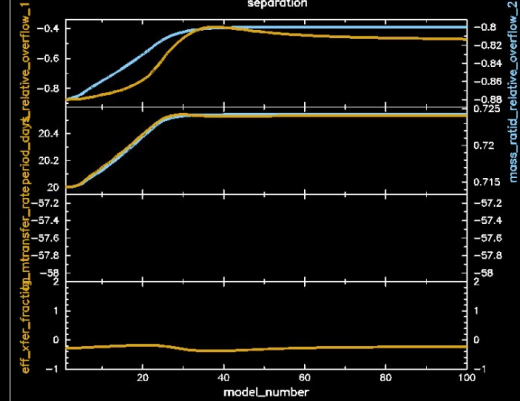
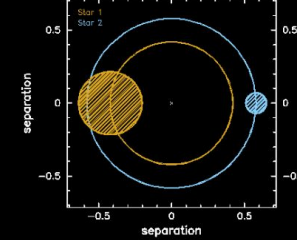
Star 2: Detached



model_number	100	log_obs_mdott	-7.2951186	star_mass	24.8500020
log_dt	1.5740304	he_core_mass	0	log_Lnuc	5.1491201
log_L	5.1486849	co_core_mass	0	log_Lneu	3.9822322
log_Teff	4.5799026	center_h1	0.3231310	log_LH	5.1491201
log_R	0.9371905	center_he4	0.6758999	log_LHe	-21.0924186
log_g	3.9590131	kh_timescale	1.193E+04	num_zones	337
surface_h1	0.7512000	mdot_timescale	4.903E+08	num_retries	0
surface_he4	0.2477736	surf_avg_omega_div_omega_crit	0.0876757		

Orbit

model 100

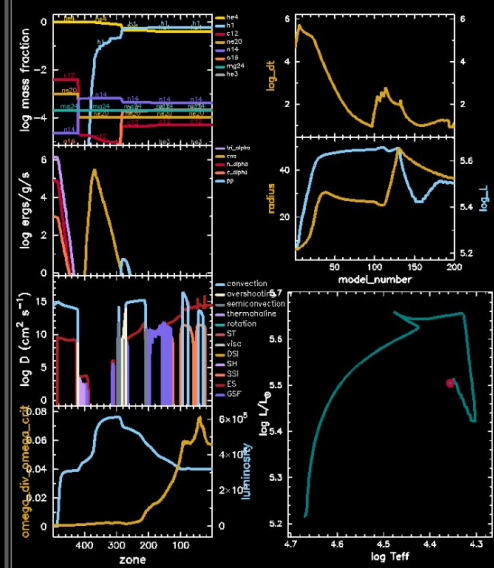


jdut	8.089E+38
mass_ratio	0.7242551
lg_mttransfer_rate	-99.0000000
period_days	20.5331248
rL_relative_overflow_1	-0.4713714
rL_relative_overflow_2	-0.7998967
eff_xfer_fraction	-0.2291263

Binaries: Thursday Lab 1

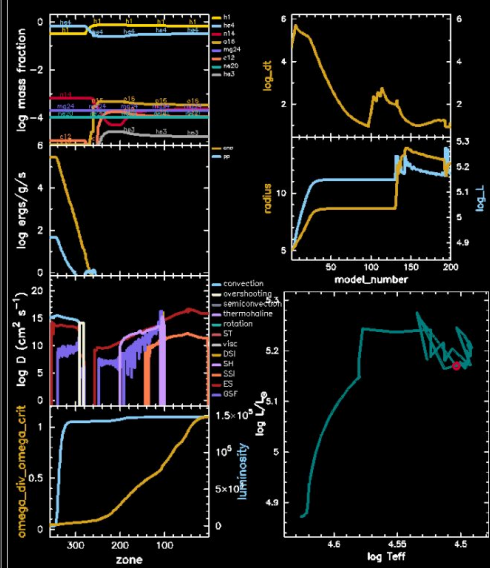
age 5.759153e6 yrs

Star 1: Donor



model_number	200	log_obs_mdott	-2.147747	star_mass	23.9588491
log_dt	1.1336698	he_core_mass	16.7601608	log_Lnuc	5.6178121
log_L	5.5046202	co_core_mass	0	log_Lneu	3.8973759
log_Teff	4.3561383	center_h1	0	log_LH	5.0622267
log_R	1.5626867	center_he4	0.9947630	log_LHe	5.4554910
log_g	2.8921602	kh_timescale	1.159E+03	num_zones	497
surface_h1	0.5996801	mdot_timescale	3.367E+03	num_retries	2
surface_he4	0.3993244	surf_avg_omega_div_omega_crit	0.0855308		

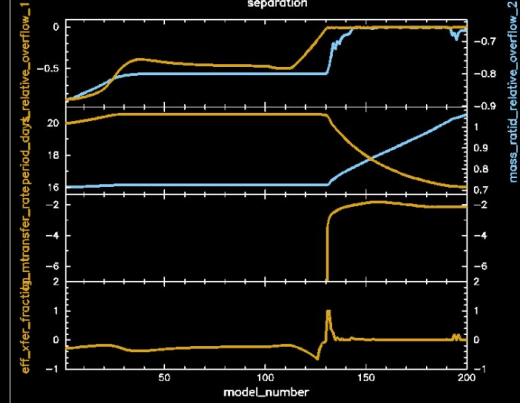
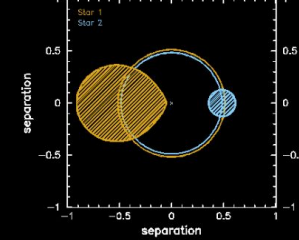
Star 2: Accretor



model_number	200	log_obs_mdott	-5.1323874	star_mass	25.4059258
log_dt	1.1336698	he_core_mass	0	log_Lnuc	5.1593771
log_L	5.1700141	co_core_mass	0	log_Lneu	3.9924849
log_Teff	4.5033076	center_h1	0.3241805	log_LH	5.1593771
log_R	1.1010450	center_he4	0.6748504	log_LHe	-21.0728100
log_g	3.6409126	kh_timescale	8.143E+03	num_zones	358
surface_h1	0.6815901	mdot_timescale	3.446E+06	num_retries	2
surface_he4	0.3173893	surf_avg_omega_div_omega_crit	0.9890574		

Orbit

model 200

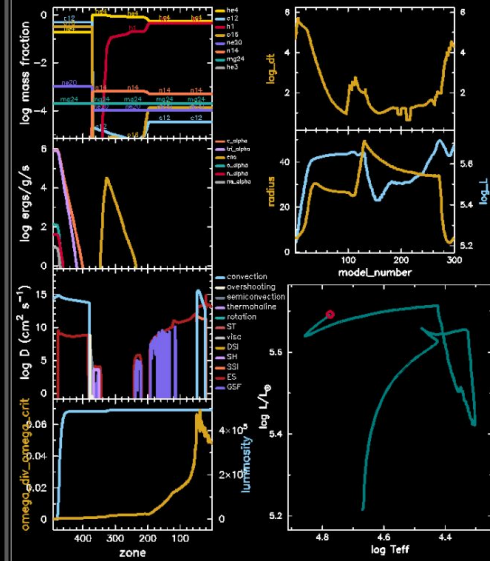


jdott	-4.351E+43
mass_ratio	1.0603984
lg_ttransfer_rate	-2.1478255
period_days	16.0250580
r_relative_overflow_1	-0.0041196
r_relative_overflow_2	-0.6650898
eff_xfer_fraction	0.0010360

Binaries: Thursday Lab 1

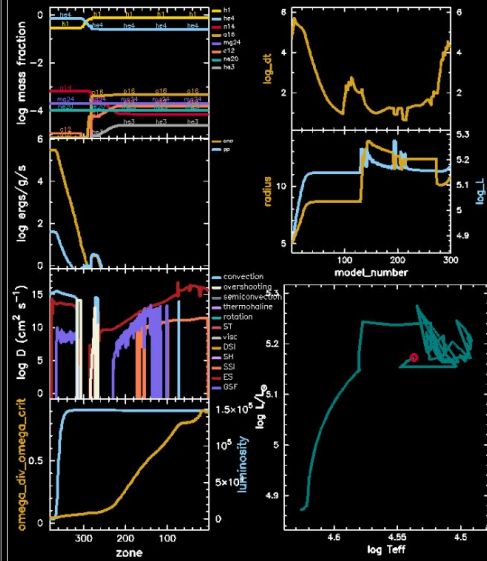
age 6.048181e6 yrs

Star 1: Detached



model_number	300	log_obs_mdott	-6.1071530	star_mass	17.9226523
log_dt	4.3019253	he_core_mass	16.9969943	log_Lnuc	5.6825742
log_L	5.6921360	co_core_mass	0	log_Lneu	2.7969425
log_Teff	4.7743830	center_h1	0	log_LH	3.7906338
log_R	0.8199552	center_he4	0.1872849	log_LHe	5.3076683
log_g	4.0515595	kh_timescale	2.326E+03	num_zones	491
surface_h1	0.4597575	mdot_timescale	2.294E+07	num_retries	5
surface_he4	0.5392566	surf_avg_omega_div_omega_crit	0.0598352		

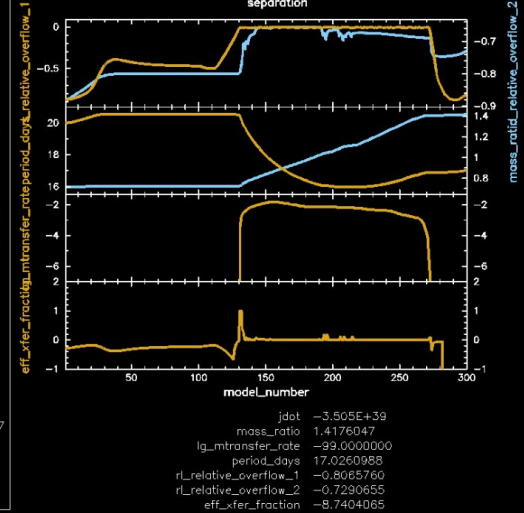
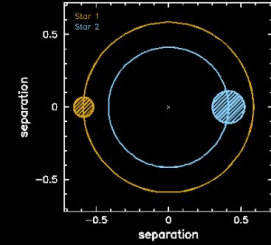
Star 2: Detached



model_number	300	log_obs_mdott	-7.0486846	star_mass	25.4072361
log_dt	4.3019253	he_core_mass	0	log_Lnuc	5.1735477
log_L	5.1730580	co_core_mass	0	log_Lneu	4.0065613
log_Teff	4.5368320	center_h1	0.2893598	log_LH	5.1735477
log_R	1.0355181	center_he4	0.7096711	log_LHe	-20.7920767
log_g	3.7719888	kh_timescale	9.404E+03	num_zones	381
surface_h1	0.7490251	mdot_timescale	2.842E+08	num_retries	5
surface_he4	0.2499487	surf_avg_omega_div_omega_crit	0.8217752		

Orbit

model 300



jdrot	-3.505E+39
mass_ratio	1.4176047
lg_mtransfer_rate	-99.0000000
period_days	17.0260988
r_relative_overflow_1	-0.8065760
r_relative_overflow_2	-0.7290655
eff_xfer_fraction	-8.7404065